

AI Agents 101: A Hands-on Introduction for Smarter Investing

20 May 2025

Mastering the future of
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What is an AI Agent?

- Agents are systems that **independently accomplish tasks on a human's behalf**.
- AI agents are systems that **perceive their environment and act autonomously**.
- AI agents are **crucial for automating tasks and enhancing efficiency** in various fields.
- Their ability to **learn from experience** makes them adaptable to new situations.
- **LLM Apps not used to control workflow execution are not Agents** e.g. LLMs sentiment classifier function.



AI Agentic Workflows Building Blocks

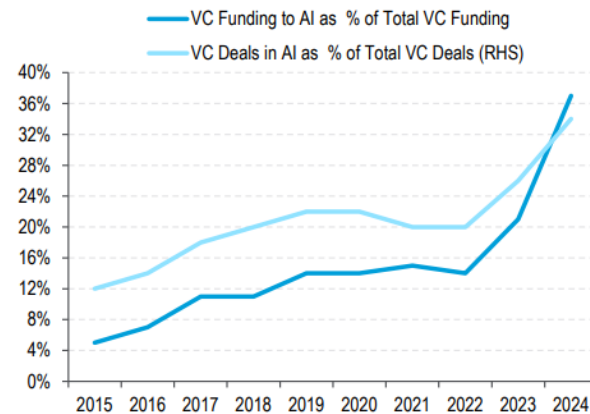
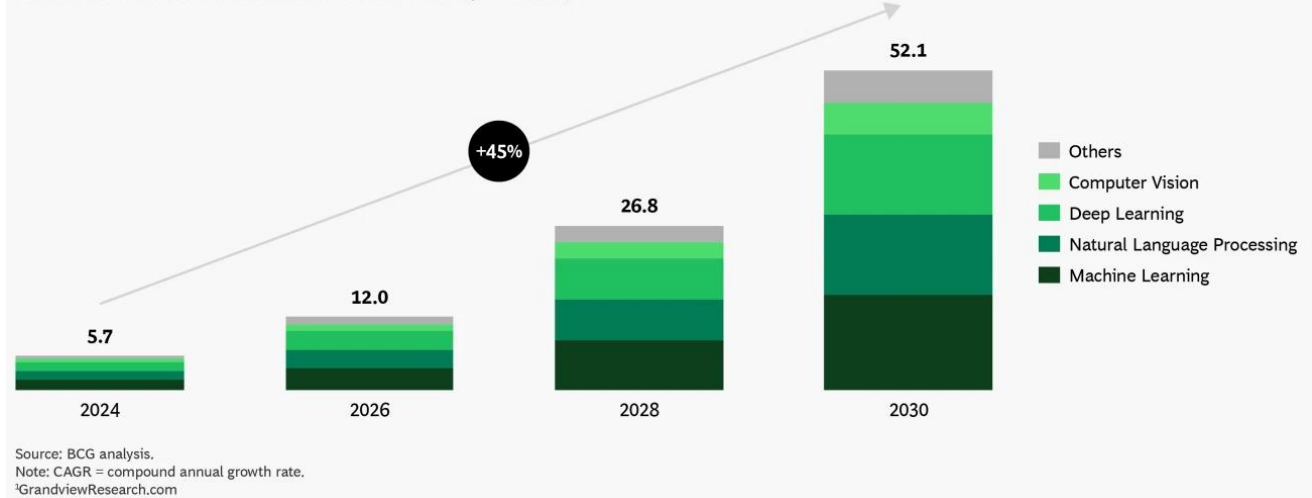
- **Augmented LLM:** key augmentations including retrieval, tools, and memory to allow the Agent to move beyond static responses and access resources (functions, tools, data) for dynamic bespoke interactions.
- **Workflow:** structured sequences of steps or actions that enable the system to achieve specific goals or solve a problem.
- **Agents:** Intelligent autonomous programs designed to attain goals via complex input understanding, reasoning, and decision-making.

AI Agents Outlook 2025-2030

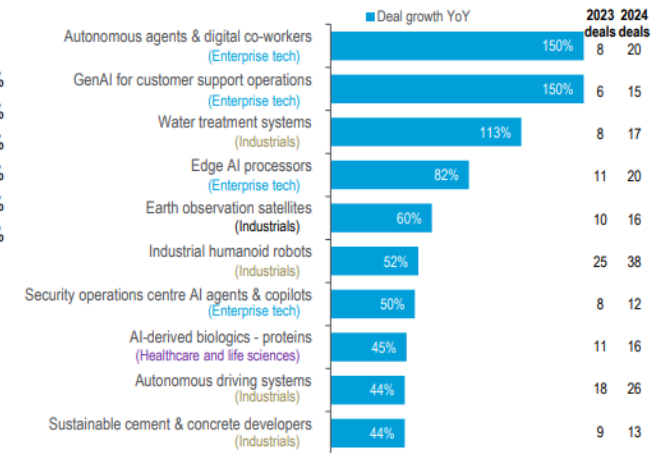
AI Agents expected to **grow at +45% CAGR** for the period 2025-2030 with several expected outcomes:

- **Augmented Humans:** AI agents integrated into teams to support workflow processes.
- **Smaller, Smarter Teams:** Complex fields (e.g. AI/ML engineering) to rely on leaner teams.
- **Transcendence vs Extinction:** corporate culture and consumer behaviour to determine if AI agents boost or replace human creativity.
- **Human Supervision:** Human supervision will be key to verify AI agents' output compliance with sensible parameters, regulatory framework and ethical rules.

AI AGENTS MARKET BY TECHNOLOGY 2024–2030¹ (\$ BILLIONS)



Source: CB Insights, Citi Global Insights



Source: CB Insights, Citi Global Insights

AI Agents in the Investments Industry

Agentic AI Use Cases	Wealth Management /Retail Banking	Corporate Banking	Institutional Investor	Insurance
 Personalized Offers	- Adaptive financial advice - Real-time savings goal optimization	- Custom lending offers - Optimized loan structures - Dynamic pricing	- Dynamic investment portfolios - Bespoke investment plans	- Tailored insurance policies - Dynamic loyalty offers
 Customer Engagement	- Virtual financial assistants - Tax and retirement planning agents	- Financial planning agents - Adaptive tax planning	- Custom research insights - Real-time market alerts	- AI-driven query handling - Proactive policy updates, renewals
 Operational Efficiency	Automate routine tasks with context-aware workflows	- Streamlined complex operations - Invoice processing, reconciliations	Automated fund review reports	Adaptive claim management
 Risk and Underwriting	- Real-time risk profiling - Predictive default modelling	Real-time risk assessment	- Diversification risk management - Real-time hedging strategies	Real-time underwriting models
 Financial Forecasting	Future savings, expense insights	Live cashflow forecasts	- Dynamic investment timing strategies - Investment performance insights	- Predicting claim reserves - Automated premium calculations
 KYC / Onboarding	- Adaptive identity verification - Real-time AML compliance	- Adaptive onboarding workflows - Real-time sanctions monitoring	- Real-time due diligence - Investor suitability analysis	Adaptive identity verification
 Fraud Prevention	Contextual-based suspicious activity detection	Corporate fraud detection	- Insider trading detection - Automated regulatory compliance	Claim fraud detection with contextual analysis

- Marketing:** A consumer goods company used AI agents to create marketing content, reducing costs by 95% and improving speed by 50x.
- Research and development:** A biopharma company reducing cycle time by 25% and gaining 35% in time efficiency.
- Customer service:** A global bank used AI virtual agents to interface with customers, reducing costs by 10x.
- Data and technology:** An IT department boosted productivity +40% by replacing legacy tech with AI agents.

Understanding AI Agents

Key Characteristics

- **Autonomous:** human intervention not required.
- **Reactive:** Responds to environment changes.
- **Proactive:** Acts to attain goals, not just react.
- **Adaptive:** Learns from experience.
- **Interactive:** collaborates with humans or other agents.

- **Simple Reflex Agents:** Act on current perceptions using condition-action rules.
- **Model-Based Agents:** Use internal models to track unobservable aspects of the environment.
- **Goal-Based Agents:** Choose actions that achieve defined goals.
- **Utility-Based Agents:** Maximize a utility function for best outcomes.
- **Learning Agents:** Improve over time by learning from past performance.

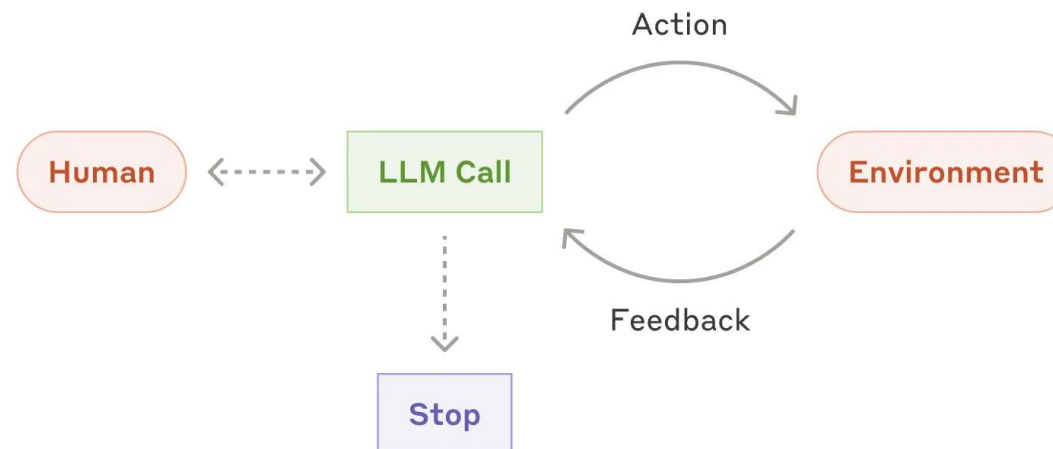
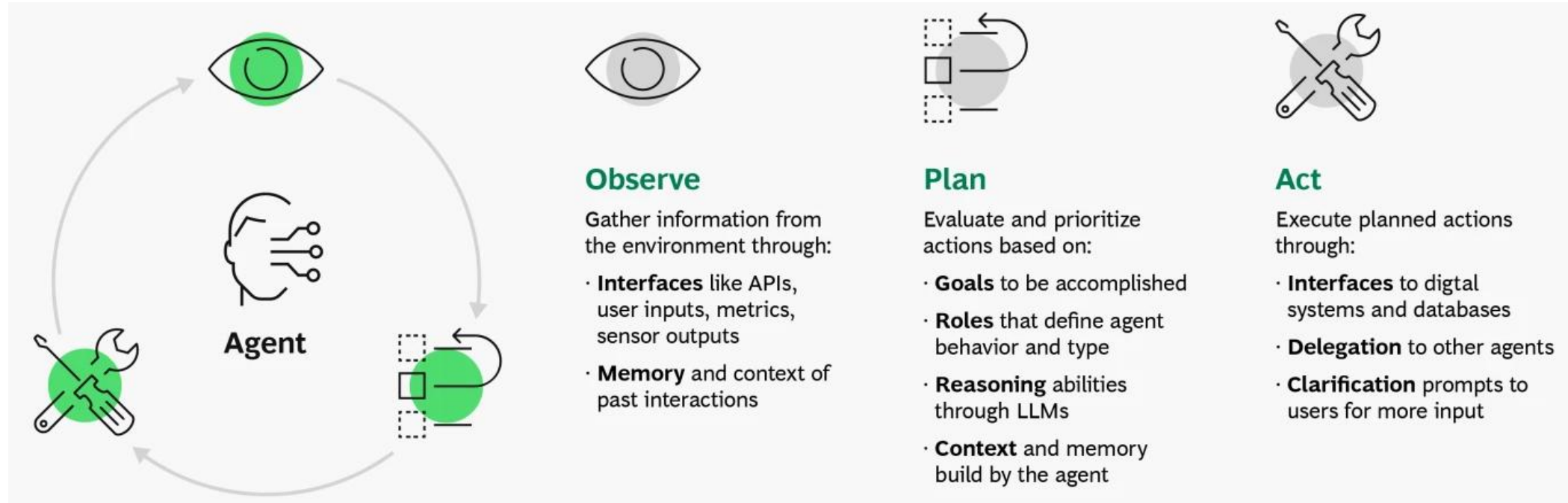
Key Type of AI Agents

AI Agents Taxonomy Evolution – Investments Focus



Agent Type	Gist	Definition	Investment Use Case	Insights
Simple Reflex Agent	Condition-Action Rules	Acts purely on current perception without memory or reasoning.	Moving Average Crossing-Price Strategy	Fast but rigid; useful for basic signal bots with no context-awareness.
Model-Based Reflex Agent	Internal Model	Maintains internal state to handle partial observability.	Track entry price and position for dynamic stop-loss	Adds basic memory; useful for risk-aware strategies.
Goal-Based Agent	Future Planning	Chooses actions to reach a predefined goal.	Minimize losses during high vol periods e.g. keep drawdown below 5%.	Enables planning; aligns strategy with investor objectives.
Utility-Based Agent	Utility Function Deployment	Selects actions to maximize a utility function (e.g., reward vs risk).	Portfolio Optimization e.g. Max Sharpe ratio	Balances trade-offs; supports nuanced decision-making.
Learning Agent	Experience Learning	Learns and improves from experience over time.	Enhance News-Driven Sentiment analyzer using stock price reaction to news headlines.	Adaptive and data-driven; ideal for dynamic or complex markets.
Hierarchical Agent	Task Decomposition	Breaks tasks into subtasks managed by sub-agents.	Decompose AAPL analysis into macro, sentiment, financials, etc	Modular design; promotes separation of concerns and explainability.
Multi-Agent System (MAS)	Collaborative/Competitive Agents Team	Multiple agents collaborate/compete in a shared environment.	Simulated Investment Team with analyst, PM, and trader.	Reflects real-world trading desks; ideal for distributed decision systems.

AI Agents Everlasting Cycle

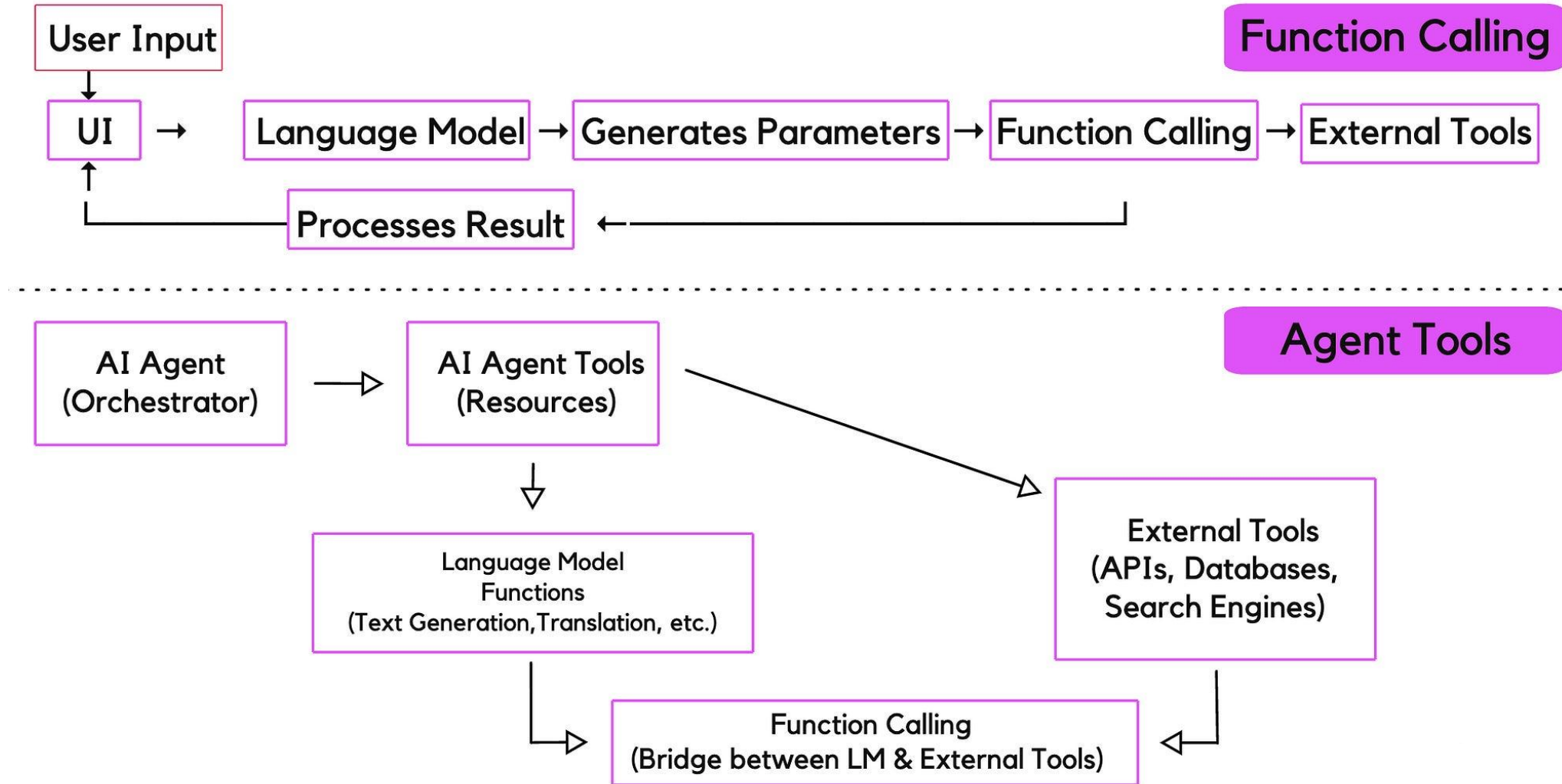


Are AI agents always the optimal solution?

- Unlike conventional automation, **Agents** are uniquely suited to workflows where traditional deterministic and rule-based approaches fall short.
- **AI Agents** are recommended for workflows with the next characteristics:

		DESCRIPTION	EXAMPLE
1	Complex Decision-Making	Workflows involving nuanced judgment, unpredictable steps, exceptions, context-sensitive decisions or no fixed-path	Regime-dependent Portfolio Optimizer accounting for client-related tax situation.
2	High Maintenance	Systems burdened by extensive, intricate rulesets that are costly or error-prone to maintain.	Multi-jurisdiction Compliance Monitoring for insider trading
3	Unstructured Data Involved	Scenarios involving natural language interpretation, document analysis, or conversational interactions.	Earnings calls Transcript/Audio analysis.

Traditional Automation vs AI Agents



Selecting Models for AI Agents

- Different **models vary in complexity, latency, and cost.**
- Use **simpler models for lightweight tasks** e.g. Retrieval, Classification.
- Reserve **more capable models for complex reasoning** e.g., Sentiment Analysis.
- **Build Agent prototype starting with the most capable** model for each task to set a performance baseline.
- **Then, try testing smaller models** to optimize efficiency without sacrificing quality

		DESCRIPTION	EXAMPLE
1	Set-Up Performance Baseline	Calculate a realistic performance baseline target factoring the financial impact of model decisions.	Human accuracy to predict stock direction from reading news is 53%-57%*.
2	Use Best Model to reach Performance Target	Build Agent prototype with the most capable model (LLM) for every task to establish a performance baseline that can be surpassed with SMLs.	The most capable model so that a smaller model can be distilled using RAG to boost accuracy, and fine-tuning for consistent behaviour.
3	Optimize Cost & Latency	Swapped into SMLs i.e. smaller specialized models to confirm performance targets are attained with similar results and lower cost and/or latency.	Earnings calls Transcript/Audio analysis.

Selecting Models for AI Agents - Example

- **Objective:** Building an Agent to conduct sentiment analysis in real-time of stock news.
- **Performance Baseline Target:** Accuracy of at least +75% vs 57% humans
- **Cost Constraints:** Spend less than \$5 per 1,000 news articles.
- **Latency Constraints:** Processing Time below 2 seconds per article.



Method	Accuracy		Cost		Latency	
gpt-4o zero-shot	71%	✗	\$1.5	✓	<1s	✓
gpt-4o few-shots (n=5)	80.5%	✓	\$9.3	✗	<1s	✓
gpt-4o-mini fine-tuned with 2k examples	80.5%	✓	\$0.9	✓	<1s	✓

- 
- **Start with gpt-4o LLM** with few shot useful to achieve accuracy target, and curate a good training set
 - **Switch to smaller model** gpt-4o-mini with fine-tuning achieves performance targets at 18% of the budgeted cost.
 - **The process is key:** sometimes a LLM will be the correct choice and others fine-tuning is not even possible due to lack of sufficient labelled samples.

Tool Kit for AI Agents

- **Tools expand an agent's capabilities** by interfacing with application/system APIs.
- **Well-documented, tested, and reusable tools:**
 - Improve discoverability
 - Simplify version management
 - Prevent redundancy

Tool Type	Definition	SDK Element	Investment Case
 Computation Tools	Perform calculations, stats, or model inferences	@tool-decorated Python function	Calculate moving average crossover or predict regime using ML model
 Data Retrieval Tools	Fetch structured data from APIs or databases	API call function or wrapper	Get real-time stock prices or news via Alpha Vantage API
 Document Tools	Read and summarize PDFs, filings, or earnings calls	Document parsers, RAG tools	Summarize Tesla's 10-K report
 Reasoning Agents as Tools	Use another agent's capabilities as a callable tool	Agent(...).as_tool()	Sentiment agent used by Portfolio Manager agent
 Actuator Tools	Perform an action in the world (place order, send alert)	Broker API, alert system	Execute/Route trade to Broker

Instructions for AI Agents

- **Instructions** act as an Agent’s system prompt **defining role, scope and behaviour**.
- **Agents use instructions for: input processing, tool deployment, and Reasoning**.
- Precise Instructions minimizes ambiguity and enhance agent decision-making, leading to a reduction in mistakes.

Instruction Type	Description	Example
 Role Definition	Set the agent's identity and expertise	"You are a technical analyst specializing in stock trends."
 Task Scope	Define what the agent should do providing smaller, clearer steps.	"Analyse moving average crossovers to generate buy/sell signals."
 Constraints & Rules	Internal Code of Conduct. Define clear and explicit actions.	"Use the pre-trained ML for technical metrics. Do not guess."
 Tool Usage Rules	If tools are registered, define when to call them.	"Use the RSI tool if momentum is unclear. Do not guess."
 Collaboration Hints	Help agent coordinate in multi-agent settings	"Use available agents as tools instead of solving everything yourself."
 Output Expectation	Describe what and how the agent should return	"Return your output as JSON with fields: ticker, signal"

Instructions for AI Agents – Example OpenAI SDK

```
# Create the agent
equity_research_agent = Agent(
    name="equity_research_agent",
    instructions="""
You are an equity research analyst.

Your task is to evaluate a company's investment potential by reviewing
available data including financials, news, and valuation indicators.

Follow this structure:
- Assign a final rating: BUY, HOLD, or SELL.
- Summarize the core investment thesis in 3-5 sentences.
- List up to 3 key risks investors should consider.
- Summarize valuation insights (e.g., P/E ratio, discount to DCF).

Always base your analysis on the provided input or data tool outputs – do not guess or hallucinate.

Output must follow the schema provided by the EquityResearchSummary model.
""",
    output_type=EquityResearchSummary,
    functions=[
        "extract_financial_summary",      # Extract financial metrics like revenue, EPS, debt
        "analyze_news_sentiment",         # Summarize and score sentiment of recent news
        "calculate_valuation_metrics",    # Compute P/E, EV/EBITDA, DCF discount
        "identify_key_investment_risks"   # Extract up to 3 critical risk factors
    ]
)
```

Clear Role Definition

Task Scope

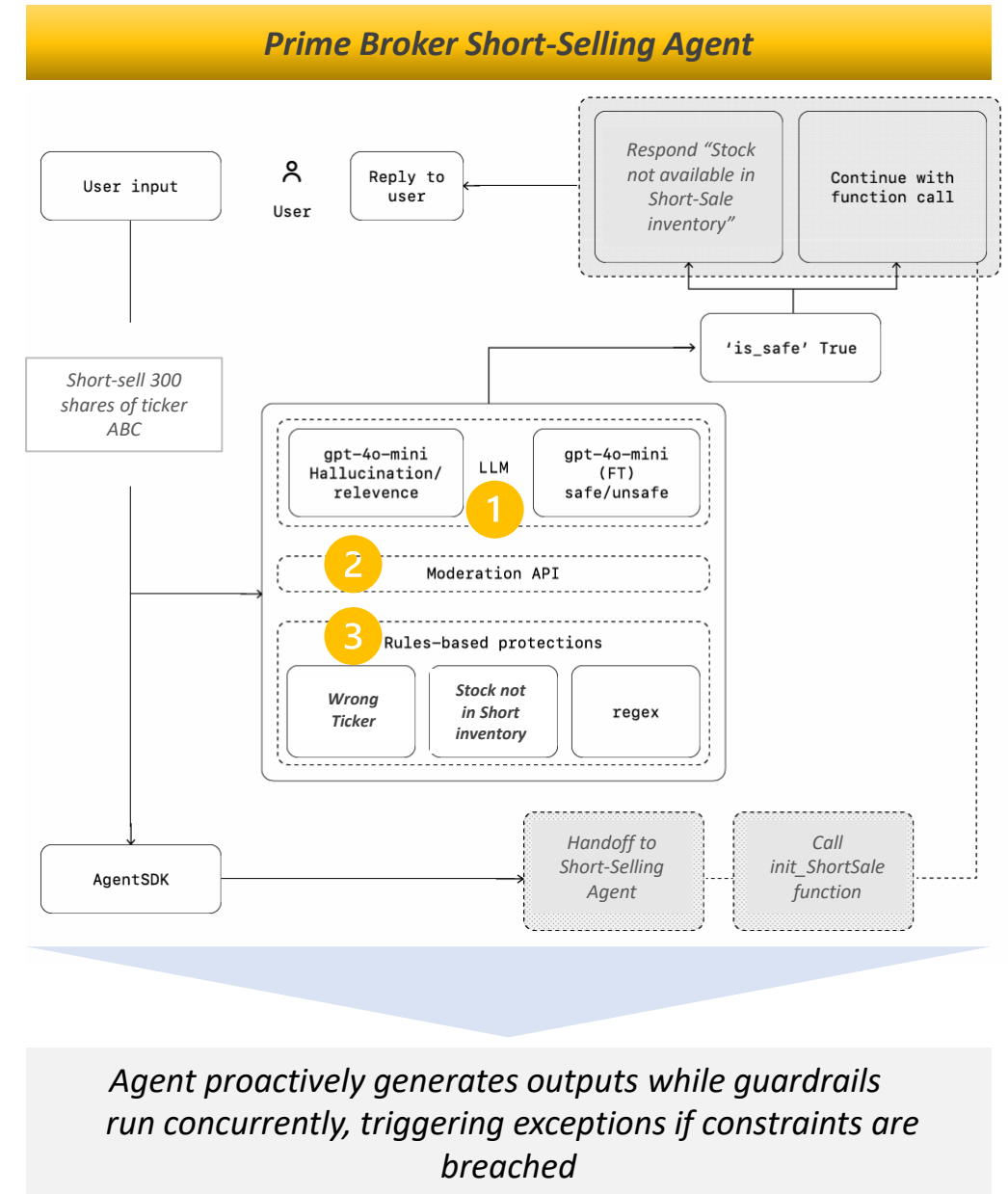
Constraints & Rules

Tools Usage + Hallucination Failsafe

Output Expectation

Guardrails

- **Guardrails** are a **defence system** to tackle risks:
 - *Output hallucination.*
 - *Safety.*
 - *Privacy.*
 - *Reputational.*
 - *Other bespoke risks.*
- Agents treat **guardrails** as first-class concepts, relying on optimistic execution by default.
- **Types of Guardrails:**
 - 1 **LLM-based:** context-aware LLM or SML to evaluate and steer outputs based on context and intent.
 - 2 **OpenAI moderation API:** Filters outputs for harmful content like hate, violence, or self-harm using OpenAI's moderation endpoint.
 - 3 **Rules-based (e.g. regex):** Applies hardcoded rules or regular expressions to validate or restrict inputs and outputs.



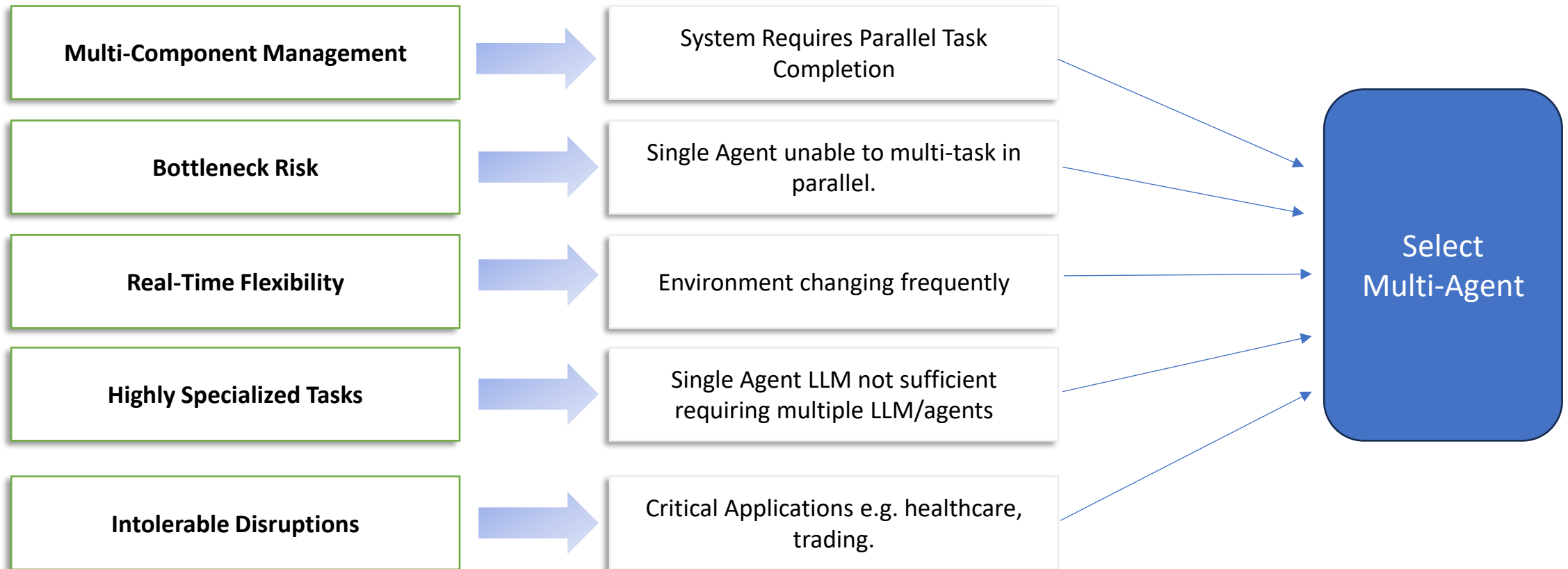
Single Agent vs Multi-Agent: Orchestration

- **Single Agent** capabilities can be boosted with tools, keeping complexity, evaluation and maintenance manageable.
- **Multi-Agent** systems can be designed for specific workflows and requirements with multiple goals.
- **Orchestration** is an important concept that depends on the type of agent system:

	SYSTEM DEFINITION	ORCHESTRATION CONCEPT
Single Agent	One autonomous agent operates independently to achieve a goal within an environment	Managing and coordinating the agent's internal components (functions, tools) and actions to achieve a specific goal
Multi-Agent	multiple autonomous agents interact or collaborate within a shared environment to achieve individual or collective goals	process of managing and coordinating multiple agents to cooperate seamlessly within a unified system to achieve complex goals

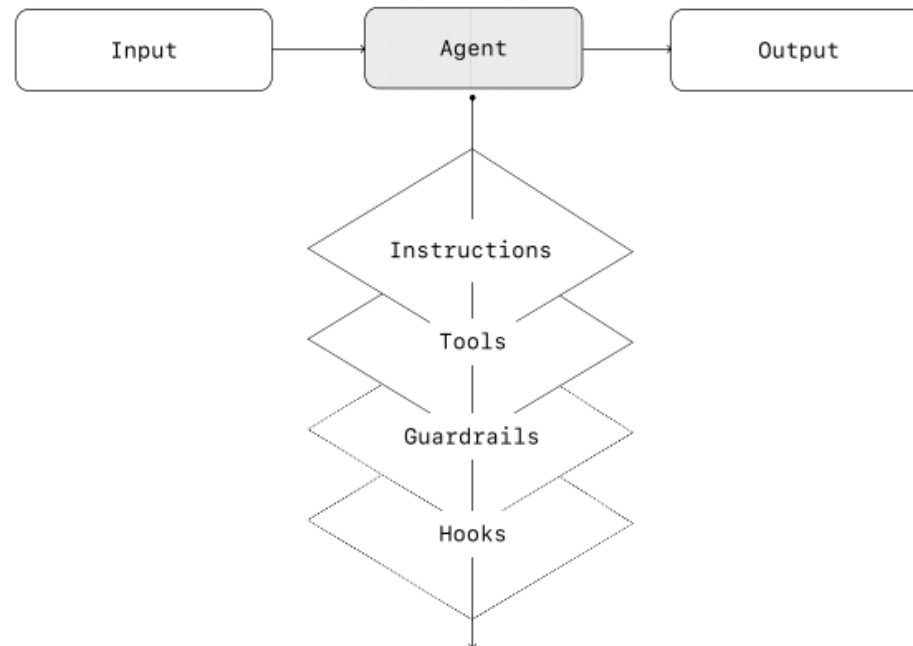
Single Agent vs Multi-Agent

- **General recommendation: start small with a Single Agent before scaling up** towards Multi-Agent systems.
- **A Single Agent may backfire** (complicated instructions, incorrect tool selection, etc): enter **Multi-Agent system**.
- For many complex workflows, splitting up prompts and tools across multiple agents improves performance.
- **Useful guidelines** can help to decide **on splitting on multiple agents**:



Single Agent Focus

- **Incremental Tooling:** Add tools to a single agent to expand capabilities while keeping complexity low.
- **Central Loop Logic:** Agents run in a loop until an exit condition is met e.g. tool call, direct output, errors, etc
- **Evaluation Made Simple:** A single flexible prompt template with policy variables can cover multiple use cases, which it's easier to maintain and adapt than many custom individual prompts for each use case.
- **Pre-Multi-Agent Strategy:** Stick with a single agent and templates before jumping to orchestration.

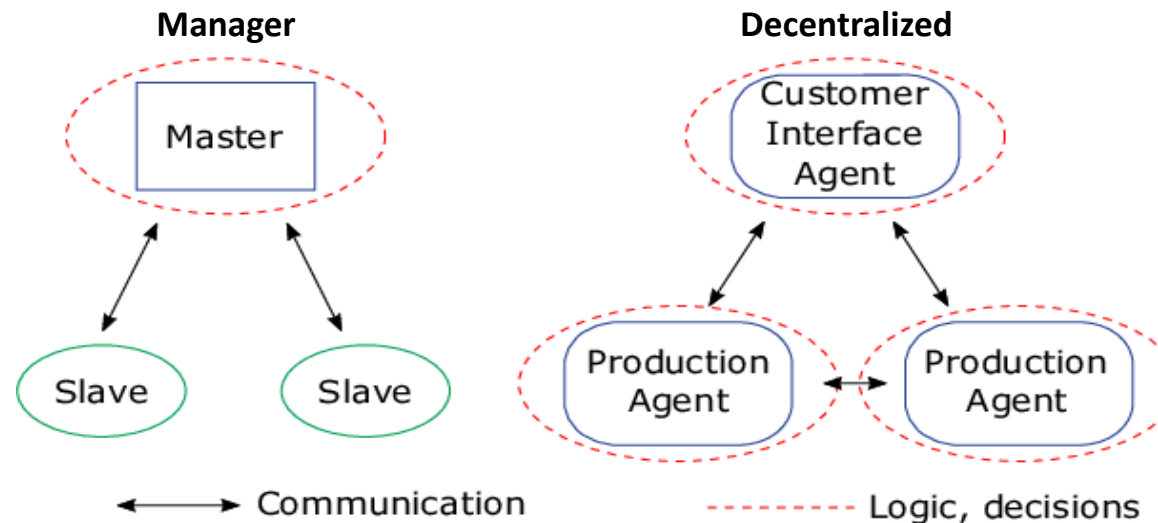


Multi-Agent Focus

- Two broad classifications for Multi-Agents Systems:

		Definition		Use When:
1	Manager	A central “manager” agent coordinates multiple specialized agents via tool calls, each handling a specific task or domain	➔	Central Agent to control workflow execution, output synthesis and user interaction is required.
2	Decentralized	Multiple agents operate as peers, handing off tasks to one another based on their specializations.	➔	Central Agent for control and synthesis not required. Agents handle execution and involve users as needed.

- Best Practices:** Keep agents modular, composable, and driven by structured, flexible prompts.



Frameworks

- **Frameworks** are 3rd party software or libraries to **help developers build, deploy, and manage AI agents**.
- **Criteria for Selecting an AI Agent Framework:**
 - Prototyping vs Production: best framework depends on whether you're building for research, production, or automation..
 - Control & Structure : A good framework should allow structured input/output, like calling APIs.
 - Observability: Tools for monitoring and debugging the agent's behaviour.
 - Flexibility: Ability to switch between LLM models/providers and integrate various models.
 - Accessibility-Control Trade-Off: Some frameworks offer more top-level abstraction, while others give granular bottom-up control.
 - Cost Efficiency: Some frameworks rely on subscription models (e.g. OpenAI APIs), which can quickly accumulate costs.
- **Debate Frameworks vs Direct LLM Usage:**
 - Some researchers favour direct LLM usage with structured prompts as more effective than using complex frameworks.
 - Other researchers believe frameworks help with organization, modularity, and reusability.
 - Scalability is key: projects destined to grow are better-off by transitioning from direct LLM usage to a more modular framework.
- **Bits & Pieces:**
 - There is no single "best" framework as it depends on the specific project needs.
 - **OpenAI SDK, CrewAI** and **LangGraph** are the most commonly used **for production**.
 - **Atomic Agents** is gaining attention as a lightweight, maintainable alternative **for research** purposes.
 - Beginners can start with simpler frameworks like **FlowiseAI** or **Phidata** before moving to complex ones.

Frameworks – Comparison Table

Framework	Definition	Best Use Case	System Type	Pros	Cons	Investment Use Case	Highlights / Expert Notes
OpenAI SDK	A Python SDK for orchestrating multi-agent workflows using OpenAI's hosted Assistants API.	Fast, reliable agent + tool workflows with GPT	Multi-Agent	Native tool calling, structured outputs, async orchestration, minimal setup	Requires OpenAI API, cloud-hosted only	Equity research pipelines, portfolio evaluation, structured investment assistants	Production-grade SDK; ideal for rapid, reliable, OpenAI-native investment systems.
CrewAI	Framework for assigning roles to agents in a "crew" to solve tasks collaboratively.	Role-based collaboration (analyst + strategist)	Multi-Agent	Easy setup, intuitive role separation, minimal code	No built-in memory, struggles with complex chains	Simulating investment analyst teams, ESG reviewers, and editorial chains	Simple, human-like agent teamwork abstraction; not ideal for multi-tool orchestration.
AutoGen	Microsoft's agent communication framework with state and retry logic.	Agent-to-agent conversations with retries, state	Multi-Agent	Flexible agent chat, supports retries, handles turn-taking	Verbose setup, not plug-and-play, more research-grade	Investment committee discussions, risk reviews, modelling regulatory dialogue	Powerful but complex; better for simulated collaborative reasoning than production chains.
LangGraph	Agent state orchestration with flow control and memory, built on LangChain.	Looped, stateful, tool-rich workflows	Multi-Agent	Graph logic, persistent memory, retry paths	Tied to LangChain, more setup required	Complex multi-step investment workflows, validation gates, due diligence pipelines	Best for custom flow control and fine-grained management; good for serious investment workflows.
Phidata	Declarative framework for building LLM tools and apps with YAML and Python.	Automation and app-like experiences	Single-Agent	Minimal boilerplate, scheduling, memory, RAG support	Less suited for deep agent collaboration	Market scanning bots, morning reports, valuation digest generation	Efficient, includes memory + RAG; great for investment task automation & data integration.
Pydantic AI	Experimental schema-safe agent execution with Pydantic typing.	Structured and auditable output	Single-Agent	Schema-enforced outputs, great for API/data alignment	No memory or orchestration, very early-stage	Generating regulation-compliant reports, KYC/AML evaluations	Strong on output structure but lacks an orchestration layer; promising but immature.
Atomic Agents	Minimalist framework for safe, structured, and testable LLM outputs.	Lightweight, modular tool-based systems	Single-Agent	Clean modularity, schema-first, easy testability	Limited orchestration tools, small community	Auditable tool-based agents for scoring, compliance, internal investment advisors	Focused on maintainability and clarity; ideal when strict structure matters.
OpenAI Swarm	Experimental concept of large-scale collaborative agent behavior (not public).	Emergent coordination and long-horizon planning	Multi-Agent	Hypothetically scalable, research-oriented collaboration	Not publicly available, speculative design	Hypothetical massive-scale portfolio monitoring, scenario simulation	Still experimental; not suited for near-term investment toolchains.
FlowiseAI + n8n	No-code / low-code platforms for LLM and API integration.	Beginner-friendly investment bots or scrapers	Single-Agent	Drag-and-drop, visual flows, built-in APIs	Limited to simple use cases, not ideal for fine control	Portfolio screener with APIs, basic news sentiment workflows	Good for non-dev teams; great entry point for prototyping low-complexity investment workflows.

Note: Please verify the table information, as some frameworks may have been updated due to the rapid advancement of AI agent technologies.

OpenAI SDK – AI Agent Lab

```
# ---- 2. Guardrail: block crypto and invalid symbols ----

class GuardrailOutput(BaseModel):
    allow: bool
    reason: str

crypto_keywords = ("bitcoin", "btc", "eth", "ethereum", "crypto", "doge", "solana", "binance", "token")

async def reject_invalid_tickers(ctx, agent, input_data):
    input_text = input_data.lower()
    if any(word in input_text for word in crypto_keywords):
        return GuardrailFunctionOutput(
            output_info=GuardrailOutput(allow=False, reason="Crypto queries are not supported."),
            tripwire_triggered=True
        )
    return GuardrailFunctionOutput(
        output_info=GuardrailOutput(allow=True, reason="OK"),
        tripwire_triggered=False
    )

# ---- 3. Compliance Agent ----

compliance_agent = Agent(
    name="Compliance Agent",
    instructions="You respond to questions that require regulatory caution. Do not use tools. Answer factually and conservatively.",
)

compliance_handoff = handoff(
    agent=compliance_agent,
    input_filter=handoff_filters.remove_all_tools # erase any previous chat info for this agent
)

# ---- 4. Trading Assistant Agent ----

trading_assistant = Agent(
    name="trading assistant",
    instructions="""
You are a stock trading assistant. You should act like a quantitative trading assistant with a professional tone.

Your Responsibilities:
- Use tools to fetch and summarize company stock data.
- Report results concisely, in markdown tables where appropriate.
- Only use retrieved data – never guess.
- Escalate to Compliance Agent if the question requests legal or regulatory advice.
- Provide BUY, SELL, or HOLD recommendation if requested.

You must follow the next rules and constraints:
- Always use tools to obtain real-time data or the provided input data. Do not guess or fabricate any numbers.
- Do not include disclaimers or additional context unless explicitly requested.
- Keep your output concise and focused – avoid verbosity, off-topic elaboration.
- If a tool fails or returns no data, clearly state this and do not make assumptions.
- Do not provide any answer to those queries that are non-investment related.

You must use markdown format:
- Use bold for headers
- Use code formatting for ticker symbols
- Use tables for numeric indicators (price, RSI, SMAs)
    """
)

tools=[get_stock_company_info, get_stock_technical],
input_guardrails=[InputGuardrail(guardrail_function=reject_invalid_tickers)],
handoffs=[compliance_handoff],
handoff_description="compliance_agent"
```



```
Q 1: What's the price trend of NVDA?
A: NVDA Price Trend

| Indicator | Value |
|---|---|
| Last Price | $135.57 |
| 50-Day SMA | $113.13 |
| 200-Day SMA | $125.92 |
| RSI | 73.04 |
| 6-Month Price Momentum | -0.08 |

- Trend Analysis:
  - The current price is above both the 50-day and 200-day SMAs, indicating a potential upward trend.
  - The RSI is above 70, suggesting the stock may be overbought.
  - The 6-month price momentum is slightly negative, indicating a recent decline in price momentum.

Q 2: What's your view on BTC?
Guardrail blocked input: Crypto queries are not supported.

Q 3: What's the valuation of Nvidia?
A: Nvidia Corporation ('NVDA') Valuation

- Current Price: $135.57
- DCF Valuation: $144.72

Company Overview:
NVidia Corporation provides graphics, compute, and networking solutions globally. It operates in two segments:
- Graphics: Includes GeForce GPUs for gaming and PCs, Quadro/NVIDIA RTX GPUs for enterprise workstation graphics, and more.
- Compute & Networking: Offers Data Center platforms, Mellanox networking solutions, and automotive AI solutions.

Sector: Technology
Industry: Semiconductors

NVidia is headquartered in Santa Clara, California, and was incorporated in 1993.

Q 4: What are the implications of EU GDPR for an asset manager?
A: The EU General Data Protection Regulation (GDPR) has significant implications for asset managers, primarily concerning how they handle personal data. Here are the key implications:

1. Data Protection Principles: Asset managers must adhere to principles such as lawfulness, fairness, transparency, purpose limitation, data minimization, accuracy, storage limitation, and integrity.
2. Lawful Basis for Processing: They need to establish a lawful basis for processing personal data, such as consent, contract necessity, legal obligation, or legitimate interests.
3. Data Subject Rights: Asset managers must facilitate and respect data subject rights, including the right to access, rectification, erasure, restriction, and portability.
4. Data Protection Officer (DPO): Depending on the scale and nature of data processing, appointing a DPO might be necessary to oversee compliance.
5. Data Breach Notification: They are required to report certain types of data breaches to the relevant supervisory authority within 72 hours and, in some cases, to the affected individuals.

Q 5: What's the market sensitivity of NVDA aka beta?
A: Market Sensitivity of NVidia ('NVDA')

- Beta: 2.114

This indicates that NVidia's stock is more volatile than the market. A beta greater than 1 suggests that the stock tends to move more than the market.

Q 6: What is a better investment NVDA or AID?
A: Investment Comparison: NVidia ('NVDA') vs. AID ('AID')

Company Overview
NVidia ('NVDA')
- Sector: Technology
- Industry: Semiconductors
- Description: Provides graphics, compute, and networking solutions.
- DCF Valuation: $144.72

AID ('AID')
- Sector: Technology
- Industry: Semiconductors
- Description: Operates as a semiconductor company with a focus on microprocessors and GPUs.
- DCF Valuation: $53.97

Technical Indicators

| Indicator | NVidia ('NVDA') | AID ('AID') |
|---|---|---|
| Last Price | $135.57 | $114.74 |
| 50-Day SMA | $113.13 | $99.04 |
| 200-Day SMA | $125.92 | $125.92 |
| RSI | 73.04 | 65.28 |
| 6-Month ROC | -0.08 | -0.17 |

Analysis
- NVidia ('NVDA') is trading above both its 50-day and 200-day SMAs, indicating a strong upward trend. The RSI of 73.04 suggests it is overbought.
- AID ('AID') is trading above its 50-day SMA but below its 200-day SMA, indicating mixed signals. The RSI of 65.28 suggests it is approaching overbought.

Recommendation
- NVidia ('NVDA'): HOLD, as it is overbought but showing strong momentum.
- AID ('AID'): HOLD, as it is approaching overbought levels with mixed trend signals.

Both companies are strong players in the semiconductor industry, but current technical indicators suggest caution due to overbought conditions.

Q 7: What is life?
A: I'm here to assist with stock trading information. If you have questions about stocks or investments, feel free to ask!
```

AI Agents: Elevating or Eclipsing Human Potential?

- Two extreme scenarios:

		GOALS	RISKS
A	Humans Displacement	Reduce labour costs and increase scalability through autonomous systems replacing humans.	Regulatory scrutiny, ethical concerns, public perception issues and security failures when AI Agents fail.
B	Humans Augmentation	Increase productivity, creativity, and efficiency without replacing humans.	Requires reskilling costs, unclear Jobs redefinition.

- A hybrid reality is the most likely scenario with some roles displaced and others augmented:
 - AI augments some roles e.g. Data Scientist, Portfolio Manager.
 - Fully automates other roles e.g. back-office operations.
 - Creates new roles e.g. Agent design, Auditing, and Orchestration.

When AI Agents Go Wrong

- AI agents often fail when faced with **limitations in reasoning, execution, or environmental context**.
- **Humans** have an **important role to play tackling AI Agents Pitfalls**:

AI Agent Failure	Core Limitation	Example/Impact	Improvement Tips
 Reasoning Failures	Hallucination, bad task breakdown, overconfident assumptions	A summarizer skips key financial insights assuming they were covered earlier	Use pydantic (structured) outputs, break tasks explicitly, and validate intermediate results
 Tool Misuse	Wrong inputs, poor tool chaining, infinite loops	A trade bot calls an API with invalid parameters, breaking execution	Add input validation, use structured output types, and limit recursion depth
 Context/Environment Gaps	No memory sharing, outdated info, API failure	Economist agent uses stale inflation data due to a broken external data feed	Introduce shared memory/state, use real-time data APIs, add retries and timeouts
 Multi-Agent Coordination	Agent conflict, overwrite, or misrouted delegation	Two agents issue conflicting buy/sell orders with no arbitration logic	Add orchestrator logic with resolution rules; apply role-based delegation
 Evaluation & Safety	No validation, prompt injection, hidden failures	Malicious prompt causes an agent to recommend high-risk investments	Sanitize inputs, implement output validators, and use trace() for full observability
 Engineering Weaknesses	Hardcoded logic, poor error handling, slow execution	Agent crashes on new asset class (e.g. crypto) with no fallback mechanism	Use async where possible, implement fallback paths, and modularize logic
 Creativity Constraints	Rigid thinking, lack of novelty unless prompted	Portfolio agent always suggests US stocks, ignoring global markets.	Frame open-ended prompts, diversify agent perspectives, and incorporate human input

The Role of Humans within an AI Agents World

- **AI Agents Pitfalls** demonstrate that **are not flawless** and require human **architects** and **supervisors**.
- **AI Agents** are still models that are **poor at thinking out of the box** when an unexpected event occurs.
- **AI Agents** will still play a **significant role as supervisors** of an AI Agent output.

Human Role	Responsibility	Investment/Trading Example
 Agent Architect	Design agent roles, tools, and task flows	Define a portfolio agent to rebalance assets and a risk agent to evaluate exposure before trades
 Evaluator & Debugger	Monitor logs, fix logic flaws, test edge cases	Review agent traces to see why a trade executed despite a stop-loss rule; debug tool misfires
 Ethical & Safety Guard	Set limits, prevent harmful actions, enforce compliance	Prevent agents from trading in blacklisted securities or during market halts
 Prompt Engineer	Tune prompts, instructions, and outputs for precision	Improve an analyst agent's prompt to better assess valuation.
 Decision-Maker/Orchestrator	Make final calls in high-stakes or ambiguous cases	Approve agent-generated trades exceeding \$1M or override during volatile market conditions

Key Takeaways:

- **AI Agent Revolution** is here to stay.
- **Human oversight remains crucial** for AI agent effectiveness.
- Future growth of **AI agents is expected to augment workforce capabilities** but **some roles will inevitably face extinction**.
- **Understanding AI agents traits is essential** for their successful deployment.
- **AI Agent Frameworks** must be selected **depending on the purpose**: research, prototyping or production.
- **Human-Machine collaboration** can lead to **innovative and efficient solutions**.



Learning Resources

- [Citi – Agentic AI Report](#)
- [OpenAI SDK Agents Intro & Documentation](#)
- [Free Course - AI Agents with LangGraph](#)
- [Free Course - Multi-Agents Systems with CrewAI](#)
- [Book - Building Agentic AI Systems. Biswas, Talukdar, 2025](#)

Carlos Salas



Carlos is an FCA-regulated Portfolio Manager, Data Scientist, and Lecturer with a proven track record in Long-only and Long/Short equity investing. He holds the CFA and CQF certifications, an MSc in Finance, and has completed postgraduate studies in Machine Learning and Data Science. Currently, he operates as an independent investment advisor and research consultant, focusing on projects that combine traditional investment frameworks with advanced data-driven and machine learning techniques. His clients include Family Offices, Ultra High Net Worth Individuals (UHNWIs), and institutional investors such as Pension Funds, Hedge Funds, and Asset Managers.

In addition to his advisory work, Carlos is deeply engaged in education and research within the investment industry. He collaborates with the CFA UK Technology Community, serves as a visiting lecturer at the University of London—where he contributes to the development of academic programs in Data Science and Machine Learning—and has previously served as a technical advisor to the IASB/IFRS Foundation, providing expert insights into the development of new IFRS accounting standards.

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What's next: Community

June 12. Skills webinar 2: Automated Fundamental
Analysis

June 19: Skills Webinar 3: AI Driven Portfolio Management

July 1: In-person networking and drinks – Pub Meetup

To register write to Aya apariy@cfauk.org

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